Battery-dependent 60 Hz signal noise

TestBoard ADC124 / Teensy 4.1
Final investigation report - 19 October 2026

Conclusion: the recordings demonstrate a strong dependence on the assembly's connection to earth. The leading explanation is externally coupled, approximately 60 Hz interference converted into signal error by high-impedance input and return paths. The precise interference source and injection point remain unmeasured.

Most decisive control	Floating battery	Battery + scope GND
No sensors: Array A median RMS [D03/D04]	0.441 mV	0.008 mV
No sensors: Array B median RMS [D03/D04]	0.711 mV	0.118 mV
Strip 2 on A: median RMS [D07/D09]	1.274 mV	0.027 mV
Strip 2 on A: PZT1 CH3 full range [D07/D09]	13.696 mV	0.806 mV

• **Grounding is an effective demonstrated intervention.** In D09, all 25 Array A channels are constant or confined to two adjacent ADC codes after warm-up.

• **Sensors amplify susceptibility but are not required.** A smaller, coherent 60 Hz pattern remains with every sensor strip and cable removed.

• **Both strip and input path matter.** Two strips tested on A and B show predominantly local increases on the connected array. B MUX1/PZT1 is especially susceptible; strip 2 magnifies that response.

• **Existing resistor groups support an impedance effect.** Without sensors, 1 Mohm groups show 2.07x (A) and 1.71x (B) the periodic RMS of balanced 470 kohm groups. This is observational, not a controlled resistor substitution.

• **Prioritize shielding and deliberate returns for untethered operation.** No resistor rework is proposed for the 0402 assembly. A new regulator or ferrite alone is not established as a remedy.

Measurement boundary: one ADC count is nominally 0.805664 mV. Sub-count RMS and fitted periodic amplitudes describe digitized code distributions; they do not independently establish analog noise below ADC resolution. Sensor mechanical health was not tested.

Report map

Setup and methods: p. 2. All recordings: p. 3. Experimental findings: pp. 4-9. Conclusions and actions: pp. 10-11. Traceability and references: pp. 12-13. Per-channel tables: pp. 14-15. Supplied schematic: p. 16.

2. Hardware, conditions and analysis

Test assembly

Laptop: ASUS ROG Zephyrus GU605CR. PCB power arrives through the laptop USB connection to Teensy 4.1. The USB 5 V feeds a 600 ohm at 100 MHz ferrite and bypass capacitors. TPS7A2033 supplies the 3.3 V component rail. REF2033 supplies 3.3 V VREF and approximately 1.65 V Vmid. In the shown schematic, ADC124S101 VA is powered from VREF, which also sets its conversion scale. [R1] [R2] [R3]

The user confirmed 60 Hz local mains and a three-pin charger plug. Initial charger comparisons used unchanged, untouched and unloaded sensors. Later tests disconnected all strips/cables, added only scope GND, or moved a single identified strip between A and B. Exact geometry and loading were not documented across every later capture. A conventional earth-referenced bench oscilloscope is assumed; charger secondary-to-earth DC continuity was not measured.

Setting	Value shared by all nine captures
Exported data	50 ADC channels: A1 = Array A; A2 = Array B
ADC scale	12 bits, nominal 3.3 V; 3300/4096 = 0.805664 mV/count
Acquisition	Both arrays; ADC scan order; blocking SPI; burst frame
SPI / timing	Requested 16 MHz SPI; repeat 1; 5 us MUX settling
Vmid sampling	Off
Processing	Filtering off; ghost removal off; raw ADC-domain CSV values

Analysis definitions

Warm-up: discard each capture's first 0.25 s. **RMS noise:** sample standard deviation around each channel's own mean, without filtering or detrending. **Array/group summary:** median of individual channel RMS values, not RMS of a pooled waveform. **Full range:** maximum minus minimum. **Robust range:** 99th minus 1st percentile.

Periodicity: native spectra identify the approximately 60 Hz fundamental and harmonics; timestamp-based least-squares fits use that fundamental plus harmonics 2-6. PCA and correlations describe shared variation but do not uniquely identify a common circuit path. For D07/D09, a common fixed 60 Hz model is used as a secondary diagnostic; raw RMS and full range are the primary matched metrics.

Timing: use CSV row times, consistent with roughly 6.53 kHz per channel. Host capture duration includes overhead and is longer than the saved signal span. Sparse gaps are preserved in timestamp fits. Figure 1 uses equal Welch settings on each capture's longest uninterrupted post-warm-up segment; spectral peak heights from unequal FFT lengths are not treated as amplitude ratios.

Final verification: all 450 primary channel RMS results in this report were recalculated from the nine original CSV files and matched to saved analysis statistics. Metadata configurations, row counts and ADC count ranges were checked. Source data were not edited.

3. Complete recording comparison

D01-D09 are report identifiers. Full source filenames appear on p. 12. Strip 1 is the first strip moved between A and B; strip 2 is identified in metadata as 6-24-26 and may contain damaged sensors. The original two-array assembly's strip identities and geometry are not fully documented.

ID	Condition	A median RMS, mV	B median RMS, mV
D01	Original sensors, battery	1.098	2.587
D02	Original sensors, charger	0.036	0.065
D03	No sensors, battery	0.441	0.711
D04	No sensors + scope GND	0.008	0.118
D05	Strip 1 on A, battery	1.017	0.636
D06	Strip 1 on B, battery	0.387	0.917
D07	Strip 2 on A, battery	1.274	0.556
D08	Strip 2 on B, battery	0.351	1.337
D09	Strip 2 on A + scope GND	0.027	0.143

Capture size and transport diagnostics

ID	Saved sweeps	Row span s	Rate / ch Hz	Reported gaps	USB frames discarded
D01	19,287	2.951	6535.2	0	2
D02	21,457	3.283	6535.9	0	0
D03	113,165	17.318	6534.4	2	27
D04	30,337	4.642	6535.0	0	4
D05	44,858	6.865	6533.8	1	14
D06	40,188	6.154	6530.4	5	33
D07	35,204	5.391	6529.6	5	33
D08	36,275	5.551	6534.3	2	9
D09	29,607	4.535	6527.4	7	37

All captures are marked complete and report zero parser rejections. Some captures contain reported transmission gaps and explicitly discarded USB frames. These sparse omissions affect timing coverage but do not explain a sustained waveform repeating approximately every 16.7 ms. Reported gap counters and the number of extended CSV row intervals are different diagnostics.

The acquisition configuration is identical across all nine files, including the final capture described as measured only on A. That CSV still contains the empty Array B as a useful secondary observation. The charger comparison and later tests were recorded at different times; historical no-strip controls do not establish identical environmental fields.

4. Charger connection suppresses pulses

D01/D02 compare the original sensor assembly on battery and with the charger attached. The user changed only charger connection. Sensor positions and board connections were unchanged for this pair.

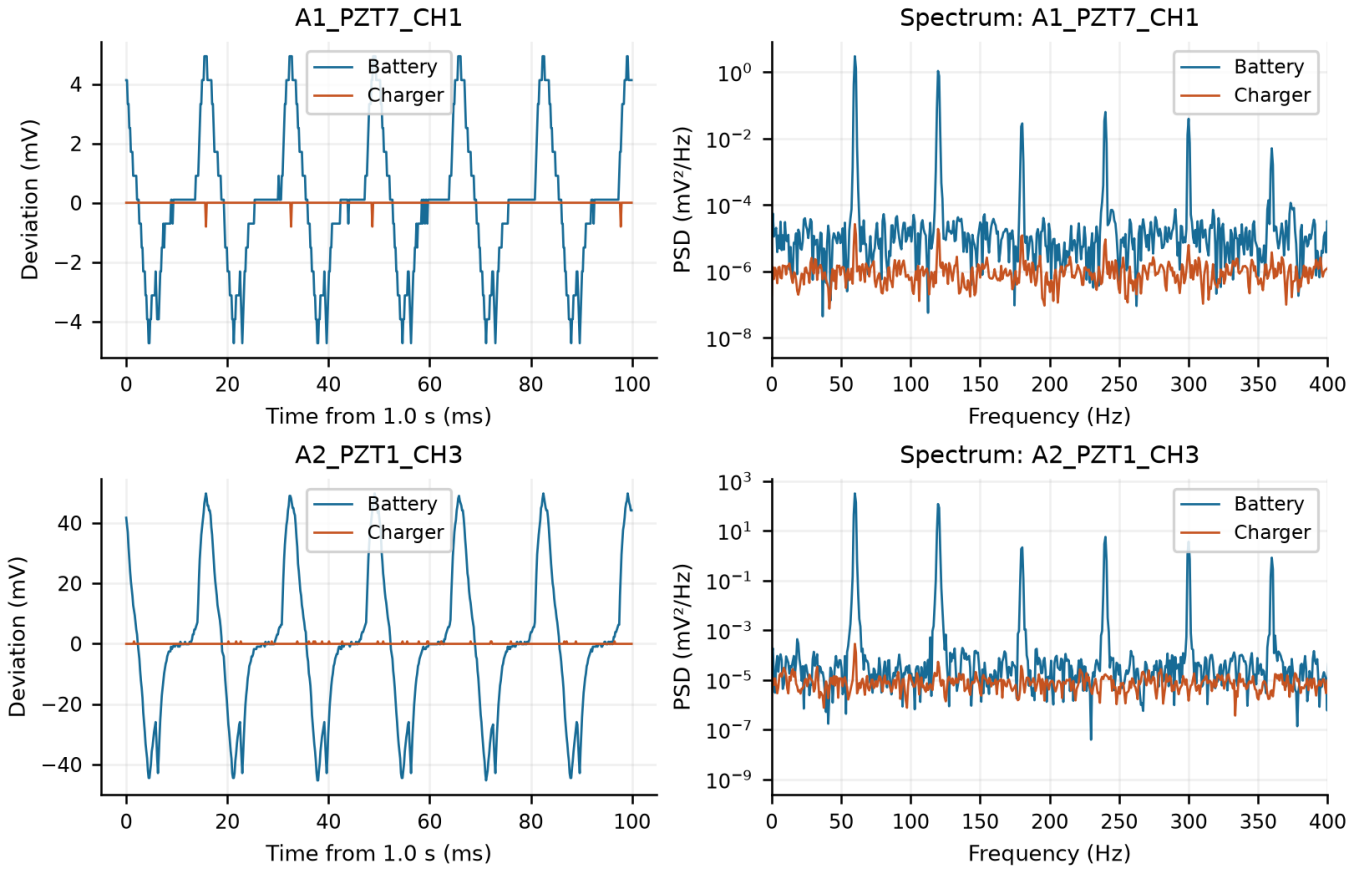


Figure 1. Original charger comparison. Left: representative 100 ms signal windows with deviations from each channel mean. Right: spectra show approximately 60 Hz and harmonics on battery, strongly reduced with charger connection. Sources: D01/D02.

Metric	Battery D01	Charger D02
A1_PZT7_CH1 RMS	2.382 mV	0.054 mV
A2_PZT1_CH3 RMS	24.430 mV	0.173 mV
A2_PZT1_CH3 full peak-to-peak	96.680 mV	1.611 mV

A timestamp-based refinement gives approximately 59.985 Hz for the battery interference; practical interpretation is approximately 60 Hz, not a calibrated last-digit frequency. Harmonics near 120, 180, 240, 300 and 360 Hz explain the pulse-like shape. The first principal component accounts for 95.4% of centered Array A variance; median absolute pair correlation is 0.890.

This pair establishes charger dependence, but alone cannot separate altered earthing, laptop power modes or USB-supply changes. The ground-only controls on pp. 5 and 9 are more discriminating because they suppress the disturbance while battery USB power remains in use.

5. Noise remains without sensors

In D03 the PCB had only the laptop USB connection: no sensors, sensor cables, charger or oscilloscope ground. The residual waveform remains approximately 59.99 Hz and is stable over the 17.3 s recording. In D04, adding only scope GND strongly suppresses that board-only disturbance.

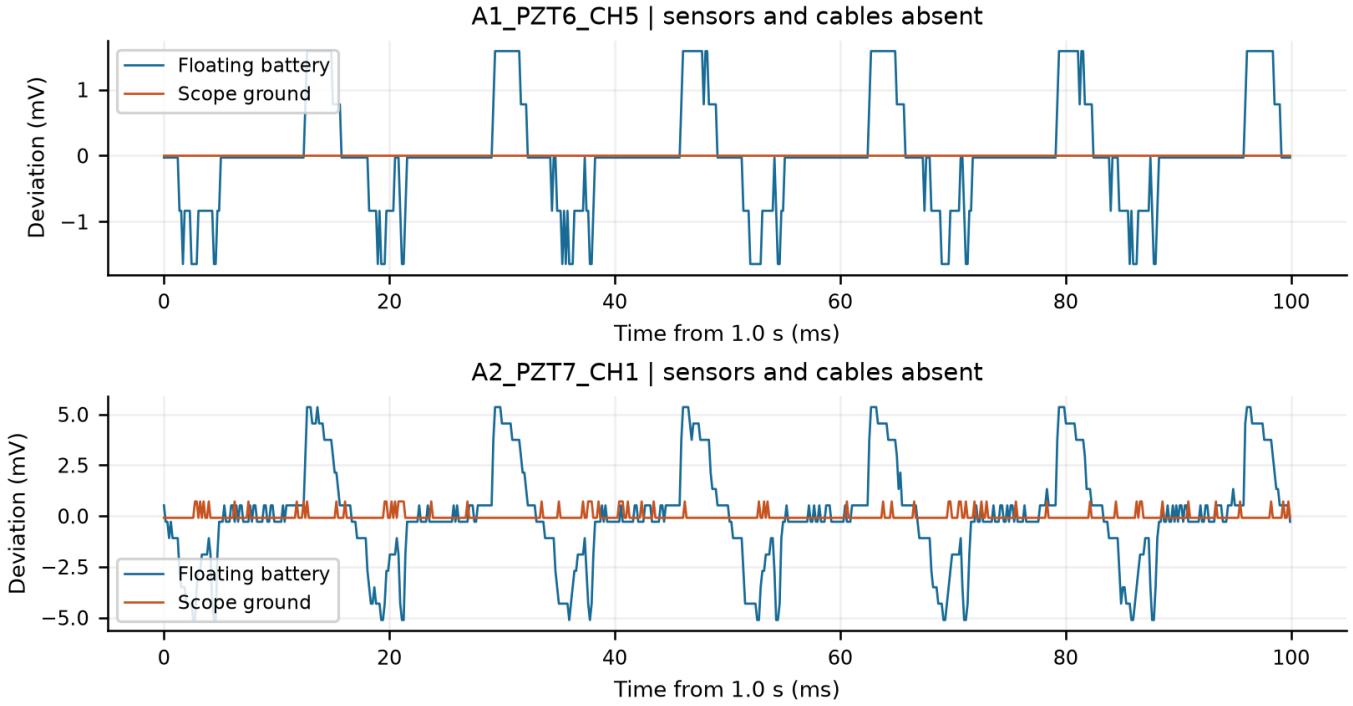


Figure 2. Sensor-disconnected floating and scope-grounded recordings at the same scale. Representative channels are the worst residual floating channels in each array. Sources: D03/D04.

Median RMS	Original sensors D01	No sensors D03	No sensors + GND D04
Array A	1.098 mV	0.441 mV	0.008 mV
Array B	2.587 mV	0.711 mV	0.118 mV

Removing the original sensor assembly reduces median RMS by 59.8% on A and 72.5% on B. The remaining disturbance is shared: the leading principal components explain 81.9% and 92.4% of centered A/B variance. Open connector pins, traces, MUX inputs, shared references and USB/common-mode paths remain in the assembly.

Conclusion: external sensors are not required for the residual interference. Their removal also changes capacitance and loading, so the reduction cannot be assigned entirely to removing an antenna. The scope-ground control strongly supports an earth-return/common-mode coupling mechanism, while the exact injection point remains unresolved.

6. Existing bias resistors affect pickup

The board uses 0402 parts and the user cannot add external resistors. This analysis instead groups channels by the existing bias resistors. Array A uses TMUX1108; the repository benchmark route model identifies Array B as TMUX1308A. Different switch types and layout limit direct causal comparisons between arrays.

Array	MUX1 / PZT1	MUX2 / PZT3	MUX3 / PZT5	MUX4 / PZT6
A	470 kohm	1 Mohm	470 kohm	1 Mohm
B	1 Mohm	470 kohm	1 Mohm	470 kohm

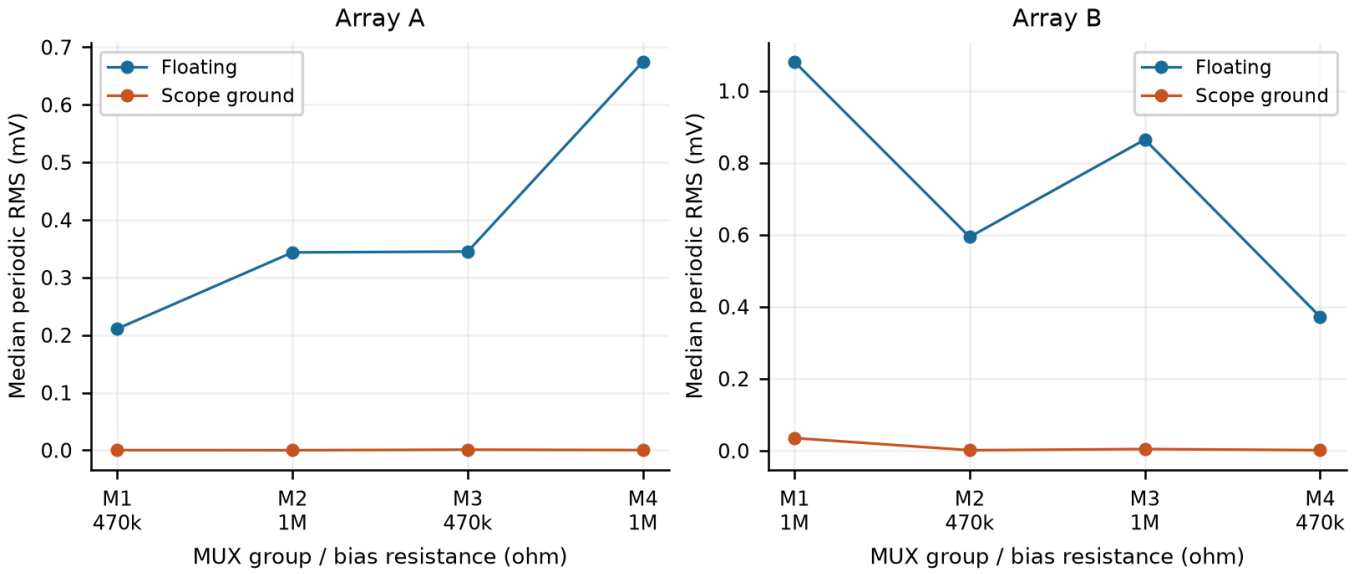


Figure 3. Physical MUX-group median fitted 60 Hz plus harmonics, sensors absent. All populated inputs are included in this plot. Sources: D03/D04; metadata routes cross-checked against the board route manifest.

Balanced no-sensor comparison	470 kohm	1 Mohm	Ratio
Array A periodic RMS	0.246 mV	0.507 mV	2.07x
Array B periodic RMS	0.526 mV	0.896 mV	1.71x

The balanced table uses addresses 0-4: five channels per physical MUX, ten per resistance per array. PZT7 is excluded from this balanced comparison because its channels span MUX1-4; it remains included in all-channel results. Only two physical MUX groups per array carry each resistance, so individual channels are not independent resistor substitutions.

For equal coupled current and resistance-dominated impedance, V = I R predicts 1,000,000/470,000 = 2.13. The observations support resistance-dependent susceptibility, but capacitance, field exposure and routing are confounded. Intrinsic resistor thermal noise is random and scales with square-root resistance; it does not explain synchronized 60 Hz pulses or their strong earthing response. [R11]

7. One strip: connected-array effect

The same physical strip was moved between A and B in D05/D06. Both arrays continued sampling, without charger, shielding or added ground. The larger noise predominantly follows the connected array; the empty array stays near the earlier no-strip baseline.

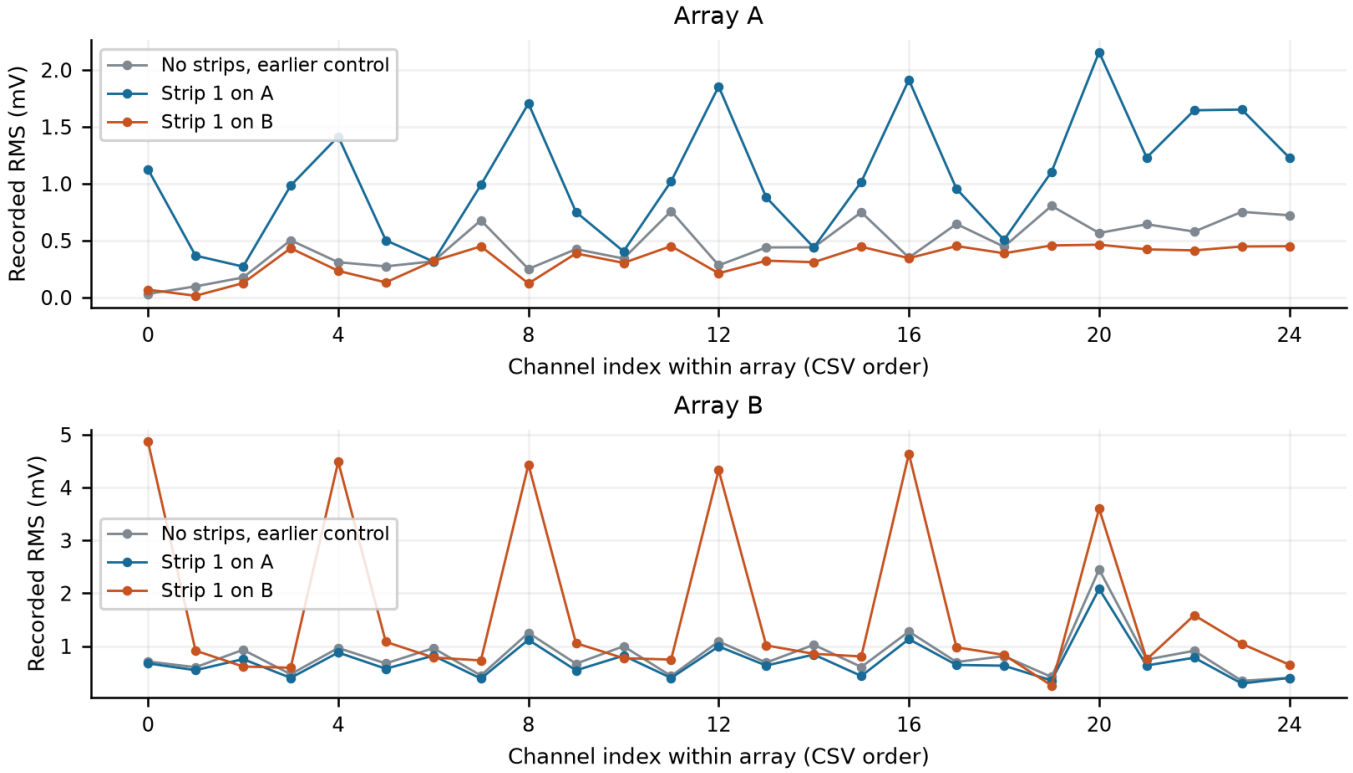


Figure 4. Strip 1 on A only and B only, compared with the earlier no-strip control. Each point is a channel RMS. The earlier baseline is not a contemporaneous environmental control. Sources: D03/D05/D06.

Connection	A median RMS	B median RMS
No strips, earlier D03	0.441 mV	0.711 mV
Strip 1 on A, D05	1.017 mV	0.636 mV
Strip 1 on B, D06	0.387 mV	0.917 mV

The disturbance remains approximately 60 Hz. Connected-array principal components explain 94.6% (A in D05) and 96.2% (B in D06) of centered variance. Cross-array synchronization is substantial, but synchronization alone is not evidence of crosstalk: both arrays can respond to the same mains field.

Array B is not uniformly worse. With the strip attached, MUX1/PZT1 median RMS is 4.463 mV on B versus 1.781 mV on A. MUX4 is quieter on B (0.731 versus 1.021 mV). MUX2 is slightly noisier on B despite its smaller resistor, demonstrating that resistance alone is an incomplete explanation.

8. A second strip magnifies the problem

D07/D08 repeat the A/B comparison using another strip, identified as 6-24-26. The user reports possible broken sensors and requests noise characterization only. Possible damage may alter loading and susceptibility; the recordings do not diagnose mechanical sensor health.

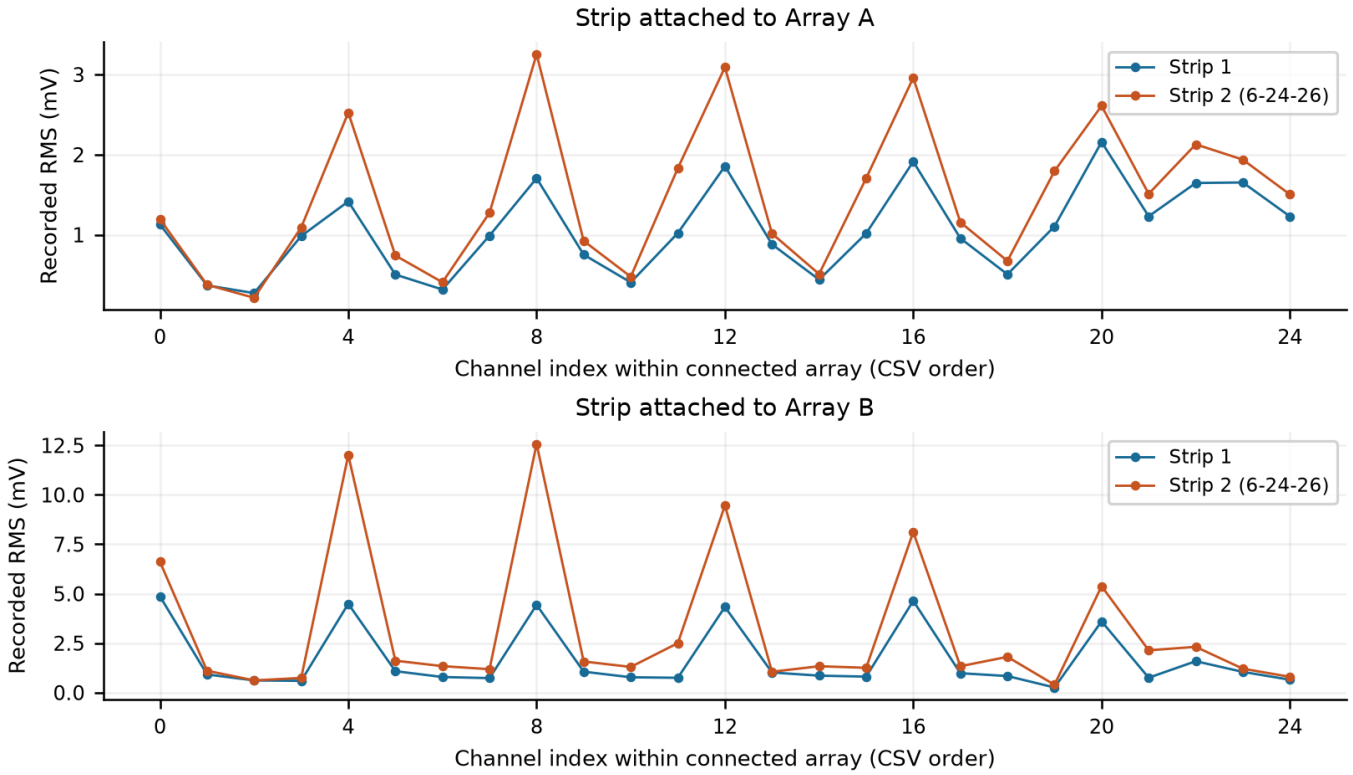


Figure 5. Strip 1 and strip 2 compared with each attached to the indicated array. Strip 2 increases noise most strongly in B MUX1/PZT1. Sources: D05-D08.

Connected-array metric	Strip 1	Strip 2
Array A median RMS	1.017 mV	1.274 mV
Array B median RMS	0.917 mV	1.337 mV
Same input: A1_PZT1_CH3 RMS	1.706 mV	3.253 mV
Same input: A2_PZT1_CH3 RMS	4.435 mV	12.573 mV

Median connected-array noise rises 25% on A and 46% on B. A2_PZT1_CH3 rises 2.83x and has a 49.951 mV full range (48.340 mV robust range). Approximately 98.5% of its variation is explained by the 60 Hz fundamental and five higher harmonics. No clipping occurs, and one-second windows show stable amplitudes.

The empty array has no corresponding increase. Both strip and input path influence the result; open electrodes, changed capacitance, leakage, return damage or geometry are possible contributors, not established faults. One strip alone reproduces pronounced B/PZT1 pickup. The larger original two-array result cannot establish a two-strip interaction because identities, geometry and timing were not controlled across those sessions.

9. Grounding suppresses strip 2 on A

D09 repeats the second-strip-on-A condition with only scope GND added to PCB GND. Acquisition settings match D07. This is an Array A control; a grounded second-strip-on-B recording has not been supplied.

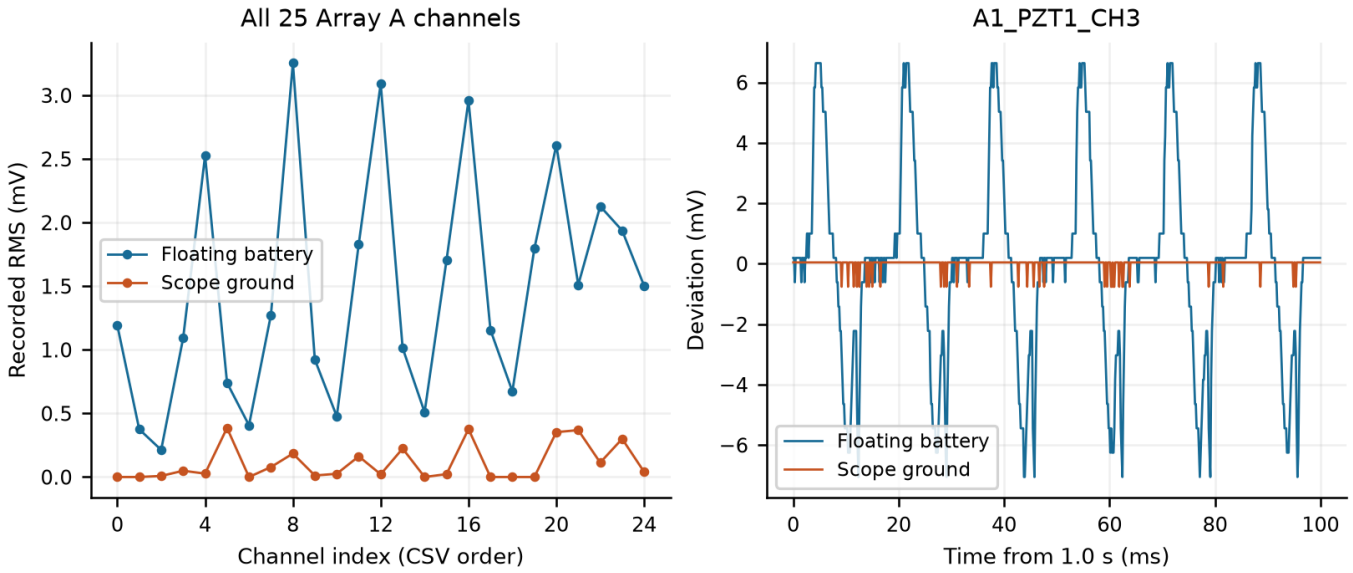


Figure 6. Matched strip-2-on-A grounding comparison. Left: every A channel. Right: the previously worst A1_PZT1_CH3 input. Sources: D07/D09.

Metric	Floating D07	Scope GND D09
Array A median recorded RMS	1.274 mV	0.027 mV
A1_PZT1_CH3 RMS	3.253 mV	0.184 mV
A1_PZT1_CH3 full peak-to-peak	13.696 mV	0.806 mV
Largest full range on any A channel	17 counts	1 count
Constant A channels after warm-up	0 of 25	7 of 25
Empty B median recorded RMS	0.556 mV	0.143 mV

The median recorded RMS reduction is 97.90% (ratio approximately 47.5). All 25 A channels are constant or limited to two adjacent codes over the analyzed 4.3 s interval. Some small periodic code-transition effects remain; this is not a claim of zero analog interference. Maximum grounded A standard deviation is 0.386 mV.

DC baselines shift little: median A mean change is 0.080 counts and maximum absolute change 0.308 counts. The principal change is suppression of AC variation. This confirms grounding sensitivity of the second strip's enhanced Array A disturbance without establishing the strip's mechanical health or quantifying the untested grounded-B case.

10. Mechanism, conclusions and limits

Working electrical model

Stage	Interpretation
External approximately 60 Hz source	Mains fields are the leading candidate because local mains are 60 Hz. A laptop display or other synchronized source is not excluded.
Unequal capacitive / return coupling	The floating laptop/PCB/strip assembly can couple differently through high-impedance signal nodes and local ground.
Conversion into differential signal error	Bias impedance, sensor capacitance, MUX state, routing and return impedance determine how common disturbance becomes ADC input error.
Earth-referenced return added	Scope GND strongly suppresses the error without replacing battery USB power or requiring charger operation.

What the circuit does and does not establish

The LDO regulates relative to local PCB ground; it does not set the complete assembly's voltage relative to earth. A 600 ohm at 100 MHz bead is not a 600 ohm impedance at 60 Hz, and a bead in the 5 V path does not block pickup directly into an input. The shown 33 ohm / 470 pF network has a simple RC corner near 10.3 MHz, not a 60 Hz rejection function. [R3] [R4]

ADC124S101 VA is both supply and reference. The shown 1 uF plus 100 nF ADC bypassing follows the manufacturer's basic recommendation. REF2033 powers VA and generates Vmid; that topology is not inherently wrong. Ideal Vmid = VREF/2 tracking can cancel reference variation in N = 4096 Vin/VREF, but unequal loading, ground impedance and settling prevent assuming ideal cancellation here. [R1] [R2]

The observations support a grounding-sensitive coupling mechanism; they do not prove the exact aggressor, rule out reference/ground disturbances, or identify a faulty ADC/LDO/reference. Shielding, PCB layout and rail waveforms were not measured. Bias-resistor comparisons remain confounded by location and capacitance. Strip 2 health remains undiagnosed.

Separate observations

DC offsets: Array B baselines remain below nominal Vmid in several conditions and change little when earthing suppresses AC noise. Those offsets require a separate settling/loading investigation. **COM4:** an earlier read-only Windows query found Teensy USB serial and Bluetooth serial both named COM4. Resolve duplicate port naming for reliable device selection, but clean parsing and a coherent grounding-sensitive waveform give little support for it as the pulse source. No port or device settings were changed.

Overall confidence: high that added scope ground suppresses the recorded disturbance; strong evidence for externally coupled/common-mode sensitivity; observational evidence that resistor value and strip identity influence susceptibility; unresolved identification of the exact physical source and entry point.

11. Suppression options and next tests

The immediate objective is clean battery operation without soldering onto 0402 parts. Actions below are proposed experiments or design options, not demonstrated remedies unless explicitly marked.

Step	Action	How to assess it
1	PCB shielding using existing GND	Use an insulated conductive enclosure/screen tied to PCB GND. Keep charger and scope ground absent. Start without sensors, then add the same strip. Compare periodic RMS and code ranges. [R7]
2	Larger local ground plate	Try an insulated plate larger than the PCB, connected to PCB GND but not earth. This changes local coupling and shielding. It is a plausible experiment, not a guaranteed piezoelectric remedy. [R8]
3	Strip / cable shielding separately	Shield interconnects with a deliberate PCB-end shield termination, keep signal and return close, and verify mechanical gain after adding capacitance. Compare PCB-only and strip-only shields independently. [R6] [R7]
4	Controlled placement and source tests	Keep strip geometry fixed. Move PCB relative to laptop/mains cables; switch off nearby lamps and unused adapters. Hold display mode fixed or turn the panel off without sleeping acquisition.
5	Second strip on B + scope GND	Repeat D08 with only scope GND added. This quantifies the largest remaining untested grounding case and helps choose the most sensitive mitigation test condition.
6	Timing and rail diagnostics	Compare 5 versus 20 us settling, logging actual rates. Measure 5 V, 3.3 V, VREF/VA, Vmid and buffer signals using differential/isolated methods that preserve the floating condition.

Filtering and future revisions

If useful sensor bandwidth permits, a 60 Hz notch plus relevant harmonic notches or a low-pass filter can reduce recorded interference. This removes genuine sensor content at those frequencies and does not establish an electrical fix. Required signal bandwidth has not been supplied.

For a future revision, evaluate intentional shield/return routing, lower impedance where sensor response allows, and buffering or balanced differential/charge-amplifier interfaces near the sensor. A lower bias resistance changes piezoelectric response and cannot be chosen from noise alone. Balanced networks and parasitic matching determine differential rejection. [R6]

If a tether is acceptable, a deliberate connection to a known instrument protective-earth terminal can reproduce the demonstrated scope-ground intervention while the laptop remains on battery. Connect only identified low-voltage PCB GND; preserve instrument protective earth. For an external power experiment, separate USB VBUS and external supply to avoid backfeeding; keeping USB ground still retains a coupling path. [R9] [R10]

12. Source catalogue and reproducibility

Every CSV below has a same-stem **_metadata.json** companion. The IDs are stable report labels, not a chronological ordering: D08 was recorded before D07. All files are dated 19 October 2026. Source directory: Documents / sensetics / data / adc on the user's computer.

ID	Original CSV filename
D01	data_no_charger_20261019_1337.csv
D02	data_with_charger_20261019_1338.csv
D03	data_on_battery_no-sensors_20261019_1433.csv
D04	data_on_battery_no-sensors_scope-gnd_20261019_1446.csv
D05	data_no-charger_sensors-array-A_20261019_1549.csv
D06	data_no-charger_sensors-array-B_20261019_1549.csv
D07	data_no-charger_sensors-array-A_20261019_1608.csv
D08	data_no-charger_sensors-array-B_20261019_1607.csv
D09	data_no-charger_sensors-array-A_ext-ground_20261019_1620.csv

Other evidence

The supplied schematic is Screenshot 2026-10-19 133644.png; it is reproduced on p. 16. The two live-display screenshots show the original noisy/quiet behavior but are qualitative context; all numerical noise metrics come from CSVs. User observations establish same-strip swaps, no external attachments in D03, scope-ground interventions, 60 Hz mains, a three-pin charger and the 0402 rework constraint.

Channel routes were read from metadata ordered_routes and checked against channel_specs labels/sample indices. Bias values follow the repository's testboard_adc124_routes.json; switch families follow Route.mux_type in testboard_adc124_benchmark.py. Both files are in Arduino_Sketches/TestBoard_ADC124/benchmarks. These assignments assume the assembled board matches those definitions. PZT7 spans physical MUX groups: CH1/MUX1, CH2/MUX2, CH3/MUX3 and CH4-5/MUX4.

Analysis archive

The sibling folder **charger_noise_investigation_20261019** contains the cumulative findings, per-channel CSV statistics, JSON summaries, plots and Python analysis scripts. The final builder is **build_final_pdf.py**. It reads the original nine CSVs, independently recalculates primary RMS values and asserts agreement with saved statistics before generating this PDF. The archive file **final_report_source_manifest.json** records file sizes and SHA-256 hashes for the 18 raw/metadata inputs.

The complete per-channel RMS tables on pp. 14-15 contain 25 A and 25 B channels across all nine conditions (450 values). This report adds no filtering to those values. Original recordings and repository code were not modified; the final PDF is the requested repository export.

13. Technical references

Primary manufacturer documentation supports the circuit interpretation and proposed remedies. These references do not establish which coupling path is active on this particular PCB. URLs checked during the 19 October 2026 investigation.

[R1] Texas Instruments, ADC124S101 datasheet
https://www.ti.com/lit/ds/symlink/adc124s101.pdf
VA is supply/reference; analog input and decoupling requirements.

[R2] Texas Instruments, REF2025/REF2030/REF2033/REF2041 datasheet
https://www.ti.com/lit/ds/symlink/ref2033.pdf
Dual VREF/VBIAS outputs, loading, tracking and capacitance limits.

[R3] Texas Instruments, TPS7A20 datasheet
https://www.ti.com/lit/ds/symlink/tps7a20.pdf
Local-ground-referenced regulation and recommended bypassing.

[R4] Texas Instruments, ferrite bead application brief SLUAAI2
https://www.ti.com/document-viewer/lit/html/SLUAAI2
Frequency-dependent ferrite impedance.

[R5] Texas Instruments, TMUX1108 datasheet
https://www.ti.com/lit/ds/symlink/tmux1108.pdf
Selected source-to-drain switching, leakage and switching behavior.

[R6] Texas Instruments, Signal conditioning for piezoelectric sensors
https://www.ti.com/lit/an/slyt369/slyt369.pdf
Sensor capacitance, charge amplifiers, bandwidth and balanced rejection of common-mode 60 Hz pickup.

[R7] Analog Devices, Electrostatic interference
https://ez.analog.com/amplifiers/a/documents/DO10677/electrostatic-interference
Capacitive interference into high-impedance inputs and shielding tied to a low-impedance local reference.

[R8] Texas Instruments, How to Remove the Ground-shift Phenomenon from Your Capacitive-sensing Application
https://www.ti.com/document-viewer/lit/html/SSZTBC0
Floating laptop/EVM example and larger local ground-plane experiment; an analogy, not a validation of this piezoelectric design.

[R9] Tektronix, ABCs of Probes Primer
https://www.tek.com/en/documents/whitepaper/abcs-probes-primer
Probe loading and grounding can change the measured circuit.

[R10] PJRC, Teensy 4.1 documentation
https://www.pjrc.com/store/teensy41.html
VIN/VUSB isolation provisions for external power.

[R11] Texas Instruments, Understanding Op Amp Noise in Audio Circuits
https://www.ti.com/lit/ab/sboa345/sboa345.pdf
Thermal resistor noise is random; voltage noise density depends on square-root resistance.

Appendix A. Array A channel RMS across all recordings

Values are nominal mV RMS (sample standard deviation). One count = 0.805664 mV. See p. 3 for conditions and p. 12 for filenames. Sub-count values describe ADC code distributions.

Channel	D01	D02	D03	D04	D05	D06	D07	D08	D09
A1_PZT1_CH1	1.329	0.339	0.032	0.000	1.127	0.068	1.194	0.010	0.000
A1_PZT3_CH1	0.353	0.058	0.097	0.000	0.368	0.015	0.375	0.034	0.000
A1_PZT5_CH1	0.256	0.030	0.176	0.000	0.271	0.126	0.213	0.075	0.008
A1_PZT6_CH1	1.034	0.000	0.503	0.000	0.983	0.433	1.091	0.411	0.049
A1_PZT1_CH2	1.649	0.400	0.310	0.196	1.415	0.234	2.525	0.270	0.027
A1_PZT3_CH2	0.502	0.000	0.273	0.000	0.503	0.132	0.741	0.103	0.386
A1_PZT5_CH2	0.317	0.010	0.318	0.106	0.316	0.320	0.407	0.315	0.000
A1_PZT6_CH2	1.111	0.084	0.678	0.012	0.989	0.449	1.274	0.441	0.078
A1_PZT1_CH3	2.036	0.215	0.250	0.005	1.706	0.123	3.253	0.134	0.184
A1_PZT3_CH3	0.730	0.384	0.425	0.391	0.751	0.387	0.923	0.387	0.012
A1_PZT5_CH3	0.407	0.000	0.342	0.064	0.405	0.304	0.476	0.289	0.025
A1_PZT6_CH3	1.084	0.008	0.759	0.000	1.021	0.451	1.831	0.438	0.162
A1_PZT1_CH4	2.211	0.025	0.282	0.000	1.855	0.212	3.091	0.210	0.022
A1_PZT3_CH4	0.812	0.036	0.440	0.000	0.880	0.323	1.014	0.290	0.224
A1_PZT5_CH4	0.437	0.000	0.441	0.041	0.439	0.310	0.509	0.298	0.000
A1_PZT6_CH4	1.098	0.000	0.749	0.062	1.017	0.446	1.705	0.421	0.023
A1_PZT1_CH5	2.275	0.073	0.350	0.267	1.913	0.345	2.959	0.343	0.377
A1_PZT3_CH5	0.922	0.000	0.646	0.323	0.956	0.452	1.156	0.434	0.000
A1_PZT5_CH5	0.532	0.336	0.446	0.000	0.506	0.388	0.673	0.351	0.000
A1_PZT6_CH5	1.176	0.000	0.805	0.000	1.105	0.457	1.795	0.447	0.000
A1_PZT7_CH1	2.382	0.054	0.567	0.008	2.158	0.464	2.609	0.446	0.353
A1_PZT7_CH2	1.280	0.000	0.644	0.000	1.230	0.423	1.510	0.399	0.368
A1_PZT7_CH3	2.012	0.050	0.579	0.076	1.646	0.414	2.128	0.376	0.113
A1_PZT7_CH4	1.757	0.402	0.753	0.402	1.652	0.448	1.937	0.437	0.300
A1_PZT7_CH5	1.400	0.099	0.723	0.353	1.229	0.450	1.503	0.416	0.040

D01 original battery; D02 charger; D03 no sensors; D04 no sensors + scope GND; D05 strip 1 on A; D06 strip 1 on B; D07 strip 2 on A; D08 strip 2 on B; D09 strip 2 on A + scope GND. All conditions still export both arrays.

Appendix B. Array B channel RMS across all recordings

Values are nominal mV RMS. Array B remains empty in D05, D07 and D09. A grounded capture of strip 2 attached to B is not present.

Channel	D01	D02	D03	D04	D05	D06	D07	D08	D09
A2_PZT1_CH1	12.683	0.076	0.711	0.398	0.675	4.866	0.628	6.615	0.402
A2_PZT3_CH1	1.800	0.358	0.602	0.045	0.547	0.917	0.505	1.104	0.052
A2_PZT5_CH1	0.951	0.060	0.932	0.154	0.756	0.615	0.695	0.616	0.143
A2_PZT6_CH1	1.412	0.167	0.470	0.212	0.401	0.593	0.380	0.741	0.214
A2_PZT1_CH2	23.073	0.065	0.967	0.254	0.880	4.490	0.814	12.010	0.278
A2_PZT3_CH2	2.961	0.092	0.674	0.012	0.574	1.082	0.529	1.613	0.014
A2_PZT5_CH2	2.432	0.237	0.963	0.013	0.820	0.786	0.736	1.337	0.014
A2_PZT6_CH2	2.340	0.057	0.447	0.066	0.388	0.731	0.365	1.182	0.071
A2_PZT1_CH3	24.430	0.173	1.252	0.016	1.122	4.435	0.993	12.573	0.015
A2_PZT3_CH3	2.837	0.284	0.669	0.015	0.544	1.057	0.475	1.571	0.012
A2_PZT5_CH3	2.364	0.196	0.999	0.366	0.821	0.774	0.740	1.297	0.375
A2_PZT6_CH3	4.901	0.041	0.436	0.010	0.400	0.747	0.372	2.500	0.005
A2_PZT1_CH4	18.371	0.031	1.090	0.384	0.996	4.331	0.874	9.463	0.353
A2_PZT3_CH4	1.782	0.055	0.690	0.359	0.634	1.015	0.556	1.053	0.344
A2_PZT5_CH4	2.301	0.110	1.027	0.118	0.843	0.856	0.743	1.332	0.163
A2_PZT6_CH4	2.224	0.000	0.607	0.170	0.437	0.806	0.389	1.252	0.178
A2_PZT1_CH5	15.724	0.017	1.277	0.367	1.134	4.639	1.056	8.119	0.353
A2_PZT3_CH5	2.343	0.000	0.701	0.262	0.646	0.980	0.577	1.332	0.238
A2_PZT5_CH5	3.513	0.335	0.813	0.013	0.631	0.836	0.554	1.807	0.008
A2_PZT6_CH5	2.587	0.006	0.418	0.007	0.353	0.257	0.313	0.395	0.005
A2_PZT7_CH1	10.088	0.354	2.452	0.254	2.084	3.601	1.869	5.380	0.190
A2_PZT7_CH2	1.496	0.046	0.748	0.051	0.636	0.751	0.540	2.138	0.030
A2_PZT7_CH3	4.641	0.398	0.914	0.069	0.783	1.586	0.722	2.312	0.061
A2_PZT7_CH4	2.837	0.013	0.345	0.046	0.294	1.046	0.290	1.208	0.043
A2_PZT7_CH5	1.664	0.021	0.401	0.402	0.401	0.650	0.403	0.795	0.403

D01 original battery; D02 charger; D03 no sensors; D04 no sensors + scope GND; D05 strip 1 on A; D06 strip 1 on B; D07 strip 2 on A; D08 strip 2 on B; D09 strip 2 on A + scope GND. Historical captures do not control all geometry and environmental differences.

Appendix C. Supplied PCB schematic screenshot

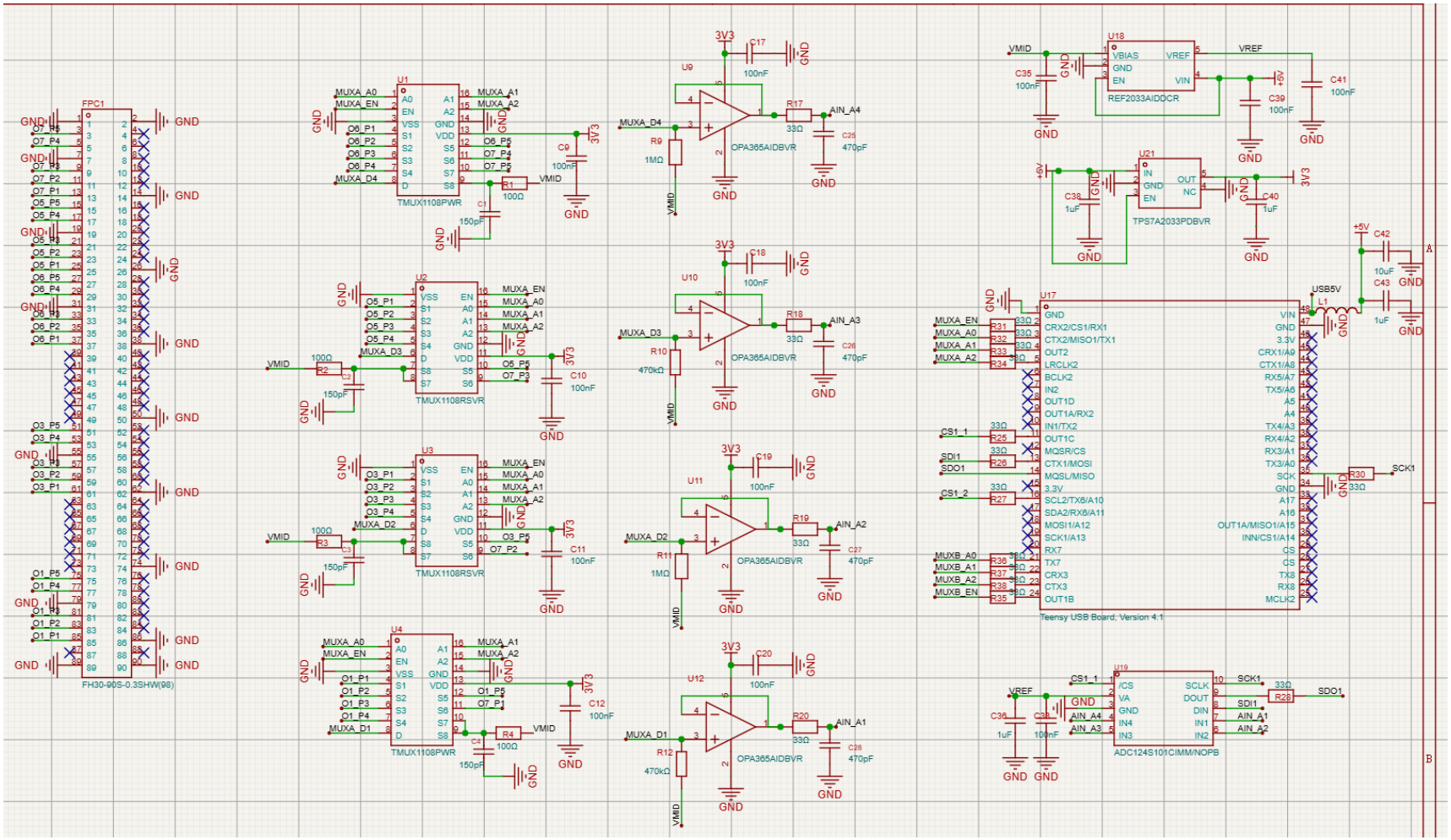


Figure 7. User-supplied schematic screenshot, retained as supplied. It shows one input bank, Teensy, the supply/reference section and ADC. Array B component/resistor mapping additionally follows the repository route manifest. Inspect the original image or native schematic for fine details; PCB layout and as-built continuity were not supplied.

Key interpretation: the ADC VA net is VREF; op-amp/MUX rail is 3V3; Vmid biases the high-resistance MUX drain/buffer input nodes. Ground symbols denote local circuit ground and do not by themselves establish protective earth. [R1] [R2] [R5]